CHAPTER 1

Question Under Discussion and Question Bias: Biased A-not-A questions in Mandarin Chinese

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1.1 INTRODUCTION

QUESTION bias, understood as a preference for one possible answer to a question over its alternatives, is an important aspect of the use conditions of interrogatives. While bias is closely connected to the semantics of questions, it is not necessarily encoded in question meaning itself. An early explicitly pragmatic treatment of this issue is offered by van Rooy & Šafářová (2003), who account for question bias in terms of the utility of possible answers, so that a question may favor one answer over another when that answer is more useful for advancing the relevant conversational goal.

This chapter develops this pragmatic perspective through a case study of *shi-bu-shi* questions in Mandarin Chinese, a type of biased A-not-A question.[1] Ye (2021a) derives the bias associated with *shi-bu-shi* questions from an asymmetry in how their possible answers contribute to resolving the Question under Discussion (QUD; Roberts 1996/2012). Building on this analysis, this chapter argues that the resulting bias is QUD-dependent, rather than directly encoded in the lexical meaning of *shi*. It further argues that question bias does not require a monopolar denotation. Although one answer may be pragmatically highlighted in context, *shi-bu-shi* questions still make both the positive and the negative answer available in their denotation.

This chapter is structured as follows. Section 1.2 sketches the bias profile of *shi-bu-shi* questions and outlines Ye's (2021a) QUD-based account. Section 1.3 examines

[1] Mandarin has several types of biased A-not-A questions; see Ye (2021b) for a utility-based account of their bias profiles. The present chapter focuses on *shi-bu-shi* questions because they offer a particularly clear case of QUD-dependent bias. I thank an anonymous reviewer for encouraging this clarification.

teleological and polar QUDs, showing that the bias in *shi-bu-shi* questions varies with the QUD. Section 1.4 turns to the relation between question bias and question denotation. By demonstrating that *shi-bu-shi* questions are bipolar despite their bias, this section argues that question bias does not provide evidence for a monopolar denotation, and that Bolinger contexts should therefore not be treated as diagnostics of question denotation. Section 1.5 concludes.

1.2 FROM MAXIMALITY TO BIAS

It has long been observed that Mandarin Chinese distinguishes two types of A-not-A questions (e.g. Tao 1998, Schaffar & Chen 2001, Shao & Zhu 2002). V-neg-V questions such as (1) are generally restricted to neutral contexts, whereas *shi-bu-shi* questions such as (2) occur in biased contexts where the questioner conjectures that the positive answer is true.

(1) ni xihuan-bu-xihuan yuyongxue?
2SG like-not-like pragmatics
'Do you like pragmatics?'

(2) ni shi-bu-shi xihuan [yuyongxue]$_F$?
2SG SHI-not-SHI like pragmatics
'Is it pragmatics that you like?'

This contrast can be further demonstrated by contextual diagnostics, following Sudo (2013). The V-neg-V question in (1) is felicitous in the neutral context in (3), but infelicitous in the biased contexts in (4)–(7). Conversely, the *shi-bu-shi* question in (2) is felicitous in the positively biased contexts in (4) and (5), but infelicitous in the neutral context in (3) or the negatively biased contexts in (6) and (7). This distribution shows that *shi-bu-shi* questions are associated with an epistemic bias towards the positive answer, while V-neg-V questions pattern as neutral questions.

(3) Neutral context (no evidence; no belief): Zhangsan is interviewing Lisi, a student he knows nothing about. He is interested in Lisi's research interests.

(4) Biased context (positive evidence; positive belief): Zhangsan is curious about Lisi's research interests. In the library, he finds that Lisi is borrowing some books on pragmatics.

(5) Biased context (no evidence; positive belief): Zhangsan and Lisi are on the phone, and they are curious about each other's research interests. Zhangsan has the impression that Lisi likes pragmatics.

(6) Biased context (negative evidence; negative belief): Zhangsan is interested in Lisi's research interests. During the interview, he finds that Lisi has never read Paul Grice.

(7) Biased context (no evidence; negative belief): Zhangsan and Lisi are on the phone, and they are curious about each other's research interests. Zhangsan has the impression that Lisi does not like pragmatics.

To account for the positive epistemic bias in *shi-bu-shi* questions, Ye (2021a) analyses *shi* as a focus marker. In particular, *shi* is truth-conditionally vacuous but carries the maximality presupposition proposed by Velleman et al. (2012), which requires its prejacent to be a possible strongest/complete answer to the current QUD, as in (8).

(8) a. $\textsc{max}_S(p) = \lambda w\,.\,\forall q \in CQ_S\,[(q >_S p) \rightarrow \neg q(w)]$
b. $[\![\textsc{shi}]\!]^S = \lambda p\,.\,\lambda w : \textsc{max}_S(p)(w)\,.\,p(w)$
c. $[\![\textsc{bu-shi}]\!]^S = \neg[\![\textsc{shi}]\!]^S = \lambda p\,.\,\lambda w : \textsc{max}_S(p)(w)\,.\,\neg p(w)$
In words: $\textsc{shi}(p)$ and $\textsc{bu-shi}(p)$ presuppose that among the possible answers to the current QUD in the context S (CQ_S), no true answer is strictly stronger than p.

To integrate the analysis with Rooth's (1992) squiggle theory, I revise Ye's (2021a) definitions in (8) as in (9), so that the contextual alternatives are compositionally accessible. Specifically, MAX, *shi*, and *bu-shi* are recast as two-place functions of the current QUD CQ and a proposition p. CQ supplies the relevant alternatives, and p is the prejacent evaluated against those alternatives.

(9) a. $\textsc{max}(CQ)(p) = \lambda w\,.\,\forall q \in CQ\,[(q \subset p) \rightarrow \neg q(w)]$
b. $[\![\textit{shi}]\!](CQ)(p) = \lambda w : \textsc{max}(CQ)(p)(w)\,.\,p(w)$
c. $[\![\textit{bu-shi}]\!](CQ)(p) = \lambda w : \textsc{max}(CQ)(p)(w)\,.\,\neg p(w)$

(10) illustrates how *shi* composes with the Rooth-style squiggle operator $\sim$. Both $\sim$ and *shi* leave the ordinary semantic value of their input unchanged, but impose presuppositional restrictions involving the current QUD. The squiggle operator presupposes that the current QUD is the subset of the focus value of IP. *Shi* then presupposes that all alternatives in that same QUD that are strictly stronger than the ordinary value of IP are false. Crucially, the first argument of *shi* is co-indexed with the contextual argument of $\sim$, ensuring that both operators make reference to the same QUD.

(10)
FocP $\rightsquigarrow \lambda w\,.\,\forall q \in \mathrm{CQ}_1\,[(q \subset [\![\mathrm{IP}]\!]^o) \rightarrow \neg q(w)]$
shi(CQ_1)
$\mathrm{IP}_\sim$ $\rightsquigarrow \mathrm{CQ}_1 \subseteq [\![\mathrm{IP}]\!]^f$
$\sim \mathrm{CQ}_1$
$[\![\mathrm{IP}]\!]^f = \{\lambda w\,.\,\text{Lisi likes } x \text{ in } w \;:\; x \in D_e\}$
$[\![\mathrm{IP}]\!]^o = \lambda w\,.\,\text{Lisi likes pragmatics in } w$
Lisi xihuan [yuyongxue]$_{\mathrm{F}}$

The maximality presupposition of *shi* eliminates stronger alternatives in the QUD set, as illustrated in Figure 1.1. In declaratives, this gives rise to an asymmetry between positive and negative sentences with respect to exhaustive inference. For instance, both (11) and (12) presuppose that stronger alternatives, such as 'Lisi likes pragmatics and syntax', are excluded. In (11), the prejacent is asserted to be true. If Lisi also likes syntax, the stronger alternative would be true, contradicting the maximality presupposition. Thus, (11) yields the exhaustive inference that Lisi does not like other branches. In (12), however, the prejacent is asserted to be false. Since other alternatives in the QUD set need not entail the prejacent, they may still be true. The sentence therefore yields the non-exhaustive inference that Lisi likes other branches.

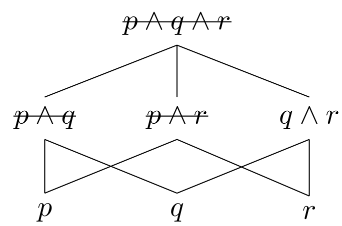

Figure 1.1 Stronger answers to the *wh*-QUD eliminated by MAX$(CQ)(p)$

(11) Lisi shi xihuan [yuyongxue]$_{\text{F}}$. (#ta ye xihuan jufaxue.)
Lisi SHI like pragmatics 3SG also like syntax
'It is pragmatics that Lisi likes. (#He also likes syntax.)'
⇝ Lisi does not like other branches.

(12) Lisi bu shi xihuan [yuyongxue]$_{\text{F}}$. (ta xihuan jufaxue.)
Lisi not SHI like pragmatics 3SG like syntax
'It is not pragmatics that Lisi likes. (He likes syntax.)'
⇝ Lisi likes other branches.

In A-not-A questions, the maximality presupposition of *shi*, projected through the Hamblin polar Q-operator as illustrated in (13),[2] yields an asymmetry between the positive and negative answers with respect to QUD resolution. The positive answer is a complete answer and can therefore close the QUD, whereas the negative answer cannot. In this sense, the positive answer is more informative and has higher utility than the negative answer.

(13)

$$[\![\text{QP}]\!] = \{\lambda w : \text{MAX}(\text{CQ}_1)(p)(w) \,.\, p(w),\ \lambda w : \text{MAX}(\text{CQ}_1)(p)(w) \,.\, \neg p(w)\}$$

$$\text{Q} \qquad \lambda r \,.\, \{r, \neg r\}$$

$$[\![\text{FocP}]\!] = \lambda w : \text{MAX}(\text{CQ}_1)(p)(w) \,.\, p(w)$$

$$\textit{shi}(\text{CQ}_1)$$

$$\text{IP}_{\sim} \quad \rightsquigarrow \text{CQ}_1 \subseteq \mathcal{A}lt(p)$$

$$\sim \text{CQ}_1$$

$$[\![\text{IP}]\!]^f = \mathcal{A}lt(p) \qquad [\![\text{IP}]\!]^o = p$$

[2] Ye (2021a) adopts Huang, Li & Li's (2009) analysis of A-not-A questions, according to which the A-not-A form morphologically realizes the Hamblin polar Q-operator, with the Q head containing a reduplicant of the initial portion of its complement and a negator. The projection of the maximality presupposition, however, does not hinge on this syntactic analysis.

Consider the context in (5), where Zhangsan is curious about Lisi's research interests and has the impression that Lisi likes pragmatics. As shown in (14), a positive answer to the *shi-bu-shi* question provides the complete answer that Lisi only likes pragmatics, thereby closing the current QUD. A follow-up question about another possible complete answer is therefore infelicitous.[3] A negative answer, by contrast, leaves the current QUD unresolved, so the questioner may felicitously ask another *shi-bu-shi* question to check a different possible complete answer.

(14) *QUD: Which branches of linguistics do you like?*

Q: ni shi-bu-shi xihuan [yuyongxue]$_F$?
2SG SHI-not-SHI like pragmatics
'Is it pragmatics that you like?'

A_1: dui.
right
'Yes, you're right.'

Q_1: #ni shi-bu-shi xihuan [jufaxue]$_F$?
2SG SHI-not-SHI like syntax
Intended: 'Is it syntax that you like?'

A_2: bu-dui.
not-right
'No, you're wrong.'

Q_2: na ni shi-bu-shi xihuan [jufaxue]$_F$?
then 2SG SHI-not-SHI like syntax
'Then, is it syntax that you like?'

Ye (2021a) further proposes the completeness-to-likelihood reasoning to derive the positive epistemic bias in *shi-bu-shi* questions. The relevant reasoning steps can be reorganized as in (15).

(15) **Completeness-to-likelihood reasoning**

a. **QUD association**: A *shi-bu-shi* question is used to resolve a larger QUD.
b. **Utility asymmetry**: Given the maximality presupposition and the relevant QUD, the positive answer to a *shi-bu-shi* question can close the QUD, whereas the negative answer cannot.
c. **Expectation bias**: Given (15-a) and (15-b), if the questioner aims to close the QUD, she expects the positive answer (i.e., $p >_{expected} \neg p$).
d. **Likelihood bias**: Given (15-c), if the questioner expects the positive answer, she should check the possible complete answer that she considers most likely to be true (i.e., $\forall q \in CQ\,[(q \neq p) \rightarrow (p >_{likely} q)]$), rather than a random one.

Essentially, the positive epistemic bias is derived from the utility asymmetry between the two answers in QUD resolution. This also accounts for the neutrality of V-neg-V questions such as (1), whose positive and negative answers are symmetric in this respect. As shown in (16), a V-neg-V question checks a possible partial answer to the current QUD. Neither the positive nor the negative answer to this V-neg-V question closes the QUD, so the questioner may ask another V-neg-V question to check a

[3] Note that this diagnostic requires the follow-up question to target the same QUD. In particular, focus must occupy the same position in the follow-up question as in the original question.

different possible partial answer. Since the two answers to the V-neg-V question have equal utility for QUD resolution, the question does not exhibit bias.

(16) *QUD: Which branches of linguistics do you like?*

Q: ni xihuan-bu-xihuan [yuyongxue]$_{\text{CT}}$?
2SG like-not-like pragmatics
'Do you like pragmatics?'

A_1: xihuan. like 'Yes, I do.'	Q_1: na [jufaxue]$_{\text{CT}}$ ni xihuan-bu-xihuan? then syntax 2SG like-not-like 'Then, do you like syntax?'
A_2: bu xihuan. no like 'No, I don't.'	Q_2: na [jufaxue]$_{\text{CT}}$ ni xihuan-bu-xihuan? then syntax 2SG like-not-like 'Then, do you like syntax?'

An important consequence of Ye's (2021a) pragmatic account is that the bias in *shi-bu-shi* questions depends on the QUD. The next section provides two pieces of evidence for this prediction. First, when the relevant QUD is teleological rather than doxastic, the modal flavor of the bias shifts accordingly. Second, in the case of a polar QUD, both answers have equal utility for QUD resolution, and the positive epistemic bias does not arise.

1.3 QUD DEPENDENCY

1.3.1 Teleological QUD

In English, the rising intonation (the L* H-H% tune) can accompany both declaratives and imperatives, as exemplified in (17). Rising declaratives may convey either positive or negative epistemic bias, whereas rising imperatives receive a suggestion-like interpretation. To provide a unified semantics for the rising intonation, Rudin (2018) proposes an extended version of the Table model that bifurcates it into doxastic and teleological halves. Declaratives interact with the doxastic half, whereas imperatives interact with the teleological half.

(17) a. Rising declaratives: *Paul was at the party?*
b. Rising imperatives: *Buy me a drink?*

Rudin (2018) also makes the generalization that a language which uses rising declaratives as biased questions also allows rising imperatives, interpreted as suggestions. This points to a broader correlation between biased-question uses and suggestion-like imperative uses of the same form. Mandarin *shi-bu-shi* questions support this correlation. As (18) and (19) illustrate, the same *shi-bu-shi* question can receive either a conjectural or a suggestion-like interpretation. In (18), the question conveys a positive epistemic bias, namely that Zhangsan may be hinting at signing him up for summer school. In (19), the question is still positively biased, but the bias is teleological rather than epistemic, yielding the suggestion-like reading that we should sign him up for summer school. This difference in modal flavor is also reflected in the choice of response particle. The conjectural reading is naturally answered with the confirmative particle *dui* 'right', whereas the suggestion-like reading is naturally answered with the acceptance particle *hao* 'okay'.

(18) *QUD_{DOX}: What is Zhangsan hinting at?*

Q: shi-bu-shi [gei ta bao yi-ge shuqi xuexiao]$_F$?
SHI-not-SHI for 3SG sign.up one-CL summer school
'Signing him up for summer school?'

A: dui.
right
'Yes, you're right.'

(19) *QUD_{TEL}: How should we improve Zhangsan's grades?*

Q: shi-bu-shi [gei ta bao yi-ge shuqi xuexiao]$_F$?
SHI-not-SHI for 3SG sign.up one-CL summer school
'Signing him up for summer school?'

A: hao.
okay
'Okay.'

The dual use of *shi-bu-shi* questions provides evidence that *shi* itself does not encode modal meaning, and should therefore not be treated as an epistemic operator (contra Wang 2024). Instead, both readings can be straightforwardly captured under Ye's (2021a) QUD-based account of bias, once Rudin's (2018) distinction between doxastic and teleological QUDs is adopted.

When the QUD is doxastic, as in (18), its possible answers can be ranked by likelihood. The *shi-bu-shi* question checks a possible complete answer to the QUD. Given the completeness-to-likelihood reasoning in (15), if the questioner aims to resolve the QUD, she should check the possible complete answer that she takes to be most likely to be true. The question therefore conveys a positive epistemic bias.

When the QUD is teleological, as in (19), its possible answers can instead be ranked by effectiveness. The *shi-bu-shi* question again checks a possible complete answer, and only its positive answer can close the QUD. Since the questioner aims to close the QUD, she expects the positive answer and should check the possible complete answer that she considers most effective. The question therefore conveys a positive teleological bias through a parallel completeness-to-effectiveness reasoning. The two types of pragmatic reasoning are generalized in (20) as completeness-to-bias reasoning.

(20) **Completeness-to-bias reasoning**

a. **QUD association**: A *shi-bu-shi* question is used to resolve a larger QUD.

b. **Utility asymmetry**: Given the maximality presupposition and the relevant QUD, the positive answer to a *shi-bu-shi* question can close the QUD, whereas the negative answer cannot.
c. **Expectation bias**: Given (20-a) and (20-b), if the questioner aims to close the QUD, she expects the positive answer (i.e., $p >_{expected} \neg p$).
d. **QUD-relative bias**: Given (20-c), if the questioner expects the positive answer, she should check the possible complete answer that she takes to be highest-ranked relative to the QUD, rather than a random one. The possible answers are ranked by likelihood under a doxastic QUD and by effectiveness under a teleological QUD, i.e., $\forall q \in CQ\,[(q \neq p) \rightarrow (p >_{likely/effective}\, q)]$.

In brief, the modal flavor of positive bias in *shi-bu-shi* questions depends on the nature of the QUD. When the QUD concerns what is true, the bias is epistemic; when it concerns what should be done, the bias is teleological. This supports Ye's (2021a) QUD-based account of bias, according to which the bias of *shi-bu-shi* questions is not intrinsically epistemic, but arises from the utility asymmetry between the two answers in resolving the QUD.

1.3.2 Polar QUD

The QUDs discussed so far have all been *wh*-QUDs. In such cases, the maximality presupposition of *shi* eliminates stronger alternatives among the possible answers to the QUD, thereby creating a utility asymmetry between the positive and negative answers to *shi-bu-shi* questions. By contrast, when the current QUD is a polar QUD, its only possible answers are p and $\neg p$, both of which are strongest/complete answers. The maximality presupposition is therefore trivially satisfied and leaves the polar QUD set unchanged. Since no utility asymmetry is introduced, Ye's (2021a) account predicts that, under a polar QUD, *shi-bu-shi* questions should not exhibit positive bias.

This prediction is borne out. In (21), for example, the prejacent contains the focused adverb *zhende* 'really'. The current QUD is a polar QUD, namely whether the addressee likes pragmatics.[4] Both the positive and the negative answers to the *shi-bu-shi* question can completely resolve this QUD. Since the two answers have equal utility for QUD resolution, the completeness-to-bias reasoning does not apply, and the question does not convey positive epistemic bias.

(21) *QUD$_{\text{DOX}}$: Do you like pragmatics?*

ni	shi-bu-shi	[zhende]$_{\text{F}}$	xihuan	yuyongxue?
2SG	SHI-not-SHI	really	like	pragmatics

'Is it really the case that you like pragmatics?'
⇝ The speaker doubts that the addressee likes pragmatics.

[4] Example (21) is infelicitous when the current QUD is a *wh*-QUD, such as *What do you like?* Thanks to an anonymous reviewer for bringing this point to my attention.

This does not mean, however, that the question is unbiased. Rather, the focused adverb *zhende* 'really' introduces negative epistemic bias, much like English *really*-questions. This bias can be explained by analyzing *zhende* in terms of verum semantics. In the literature, there are two main approaches to verum in polar questions, namely the focus approach (e.g., Goodhue 2018) and the operator approach (e.g., Romero & Han 2004). Either approach can in principle derive the negative epistemic bias observed in the *shi-bu-shi* question in (21). For present purposes, I follow the general insight of Bill & Koev (2021) and propose (22) as a preliminary lexical entry for *zhende* 'really'. Its ordinary semantic value is the identity function, with a presupposition that the prejacent is not straightforwardly acceptable.[5] Its focus semantic value is a set containing the identity function and the function mapping a proposition to its negation.

(22) Semantics of *zhende* 'really'

a. $[\![zhende]\!]^o = \lambda p : p \text{ is not straightforwardly acceptable} \,.\, p$

b. $[\![zhende]\!]^f = \{\lambda p \,.\, p,\ \lambda p \,.\, \neg p\}$

Given the presupposition in (22-a), the prejacent of *zhende* 'really' must meet contextual resistance, which may come from contextual evidence, expectations, or conversational assumptions. Thus, when *zhende* occurs in *shi-bu-shi* questions, the prejacent is raised as one member of a polar alternative set, but against a background in which accepting it is contextually non-trivial. In other words, the positive answer to the *shi-bu-shi* question is not straightforwardly acceptable, which gives rise to negative epistemic bias.

(23)
- QP
  - Q: $\lambda r \,.\, \{r, \neg r\}$
  - FocP $\rightsquigarrow \textsc{max}(\mathrm{CQ}_1)(p)$
    - *shi*(CQ_1)
    - $\mathrm{IP}_{\sim}$ $\rightsquigarrow \mathrm{CQ}_1 \subseteq \{p, \neg p\}$
      - $\sim \mathrm{CQ}_1$
      - $[\![zhende_{\mathrm{F}}\ \mathrm{IP}]\!]^f = \{p, \neg p\}$
$[\![zhende_{\mathrm{F}}\ \mathrm{IP}]\!]^o = p$
$\rightsquigarrow p$ is not straightforwardly acceptable

In sum, polar QUDs show that the bias associated with *shi-bu-shi* questions is not directly encoded in the lexical meaning of *shi*. When the positive and negative answers have equal utility for QUD resolution, the positive bias normally associated with *shi-bu-shi* questions does not arise, as predicted by Ye's (2021a) QUD-based account. The negative bias found in examples with *zhende* 'really' therefore has a different source. As a verum-like item, *zhende* indicates that accepting the prejacent is not straightforward in the context, thereby giving rise to negative epistemic bias.

[5] In terms of the Table model (Farkas & Bruce 2010), this presupposition requires that the prejacent not already be settled as a projected common-ground update in the context.

This shows that question bias may arise from distinct sources, including QUD-based reasoning and the contribution of particular lexical items.

1.4 BIAS WITHOUT MONOPOLARITY

In Hamblin's (1973) classic analysis, polar questions denote a bipolar set, $\{p, \neg p\}$. This analysis, however, faces a challenge in capturing bias variation among polar questions. Krifka (2017, 2022) addresses this challenge by arguing that at least some polar questions are monopolar, in the sense that they present only one proposition to the addressee. More recently, Matthewson (2025) uses certain biased contexts identified by Bolinger (1978), which I refer to here as Bolinger contexts, as a diagnostic for monopolarity.

The preceding discussion has treated the bias in *shi-bu-shi* questions as a pragmatic effect, derived from QUD-based reasoning and, in some cases, from the contribution of particular lexical items. This raises a further question about the relation between question bias and question denotation: Does a biased question have to be monopolar, or can it nevertheless retain a bipolar denotation? In what follows, I first demonstrate that *shi-bu-shi* questions are bipolar in their denotation, and then show that they can nevertheless occur in Bolinger contexts. This suggests that question bias is not sufficient evidence for a monopolar denotation, and that Bolinger contexts should therefore not be treated as diagnostics for question denotation.

First, *shi-bu-shi* questions can be embedded by the responsive predicate *zhidao* 'know', and give rise to Q-to-P distributivity inferences (cf. Theiler, Roelofsen & Aloni 2019). For instance, (24) yields the inference that Zhangsan knows Lisi likes pragmatics or Zhangsan knows Lisi does not like pragmatics. This shows that *shi-bu-shi* questions are bipolar in their denotation, since such an inference requires both the positive and the negative answer to be available. A monopolar question would not generate this kind of Q-to-P distributivity inference.

(24) Zhangsan zhidao Lisi shi-bu-shi xihuan yuyongxue.
Zhangsan know Lisi SHI-not-SHI like pragmatics
'Zhangsan knows whether Lisi likes pragmatics.'

Second, *shi-bu-shi* questions can also occur in the antecedent of unconditionals, where they produce another type of Q-to-P distributivity inference. For instance, (25) yields the inference that if Lisi likes pragmatics, you have to invite him, and if Lisi does not like pragmatics, you have to invite him. This provides further evidence that *shi-bu-shi* questions make both polar alternatives available in their denotation, since the observed distributive inference requires quantification over both the positive and the negative answer.

(25) wulun Lisi shi-bu-shi xihuan yuyongxue, ni dou dei qing ta.
no.matter Lisi SHI-not-SHI like pragmatics 2SG DOU have.to invite 3SG
'No matter whether Lisi likes pragmatics or not, you still have to invite him.'

Yet *shi-bu-shi* questions are felicitous in Bolinger contexts, where the prejacent is the most likely alternative while other alternatives remain possible, as shown in (26). Matthewson (2025) takes such contexts to diagnose monopolarity. The behaviour of *shi-bu-shi* questions, however, suggests that this conclusion is too strong. As shown by the embedding and unconditional facts above, *shi-bu-shi* questions have a bipolar denotation despite their bias. Their felicity in Bolinger contexts therefore cannot be taken to show that only the positive answer enters the denotation. Rather, it shows that the positive answer is pragmatically highlighted in the context. Question bias is therefore not sufficient evidence for a monopolar denotation. Bolinger contexts diagnose which answer is contextually highlighted, and hence bear on question bias rather than on question denotation.

(26) Context: Mary and Bella run into each other and Bella looks tired. Mary asks, "Are you ok?" Bella replies, "I'm so tired. I can't sleep in the same room as my husband any more." Mary asks:

ta shi-bu-shi dahulu?
3SG SHI-not-SHI snore
'He snores?'

1.5 CONCLUSION

Building on Ye's (2021a) pragmatic analysis, this chapter has argued that the bias associated with *shi-bu-shi* questions is QUD-dependent, rather than directly encoded as a fixed component of the lexical meaning of *shi*. The main evidence comes from teleological and polar QUDs. When the relevant QUD is teleological, the modal flavor of the bias shifts accordingly. When the current QUD is polar, the positive bias normally associated with *shi-bu-shi* questions does not arise, since the positive and negative answers have equal utility for QUD resolution. The negative epistemic bias found in examples with *zhende* 'really' has a different source, namely the verum-like contribution of *zhende* itself.

This chapter has also argued that question bias does not require a monopolar denotation. Evidence from embedding under *zhidao* 'know' and from unconditionals shows that *shi-bu-shi* questions make both positive and negative answers available in their denotation. Their felicity in Bolinger contexts therefore cannot be taken to show that only the positive answer enters the denotation. Rather, such contexts identify the answer favored by the bias. More generally, *shi-bu-shi* questions show that question bias and question denotation should not be conflated, since a question may be biased while retaining a bipolar denotation.

References

Bill, Cory & Todor Koev. 2021. Verum accent IS VERUM, but not always focus. In Patrick Farrell (ed.), *Proceedings of the Linguistic Society of America*, vol. 6, 188–202. LSA. https://doi.org/10.3765/plsa.v6i1.4959.

Bolinger, Dwight. 1978. Yes-no questions are not alternative questions. In Henry Hiż (ed.), *Questions*, 87–105. Dordrecht: D. Reidel Publishing Company.

Farkas, Donka F. & Kim B. Bruce. 2010. On reacting to assertions and polar questions. *Journal of Semantics* 27(1). 81–118. `https://doi.org/10.1093/jos/ffp010`.

Goodhue, Daniel. 2018. *On asking and answering biased polar questions*. McGill University dissertation.

Hamblin, Charles L. 1973. Questions in Montague English. *Foundations of Language* 10(1). 41–53. `http://www.jstor.org/stable/25000703`.

Huang, C.-T. James, Y.-H. Audrey Li & Yafei Li. 2009. *The syntax of Chinese*. New York: Cambridge University Press.

Krifka, Manfred. 2017. Negated polarity questions as denegations of assertions. In Chungmin Lee, Ferenc Kiefer & Manfred Krifka (eds.), *Contrastiveness in information structure, alternatives and scalar implicatures* (Studies in Natural Language and Linguistic Theory 91), 359–398. Cham: Springer International Publishing Switzerland.

Krifka, Manfred. 2022. Modelling questions in Commitment Spaces. In Moritz Cordes (ed.), *Asking and answering: Rivalling approaches to interrogative methods*, 63–95. Tübingen: Narr Francke Attempto.

Matthewson, Lisa. 2025. Polar questions in n e kepmxcín: monopolar, bipolar, and exhaustive. *Natural Language Semantics*. `https://doi.org/10.1007/s11050-025-09239-6`.

Roberts, Craige. 1996/2012. Information structure: towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5(6). 1–69. `https://doi.org/10.3765/sp.5.6`.

Romero, Maribel & Chung-Hye Han. 2004. On negative *yes/no* questions. *Linguistics and Philosophy* 27(5). 609–658. `https://doi.org/10.1023/B:LING.0000033850.15705.94`.

Rooth, Mats. 1992. A theory of focus interpretation. *Natural Language Semantics* 1(1). 75–116. `https://doi.org/10.1007/BF02342617`.

Rudin, Deniz. 2018. *Rising above commitment*. University of California, Santa Cruz dissertation.

Schaffar, Wolfram & Lansun Chen. 2001. Yes-no questions in Mandarin and the theory of focus. *Linguistics* 39(5). 837–870. `https://doi.org/10.1515/ling.2001.036`.

Shao, Jingmin & Yan Zhu. 2002. *Shì-bù-shì VP* wenju de kendingxing qingxiang ji qi leixingxue yiyi [The affirmative inclination of the *shi-bu-shi VP* question and its typological significance]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]* 3. 23–36.

Sudo, Yasutada. 2013. Biased polar questions in English and Japanese. In Daniel Gutzmann & Hans-Martin Gärtner (eds.), *Beyond expressives: Explorations in use-conditional meaning* (Current Research in the Semantics/Pragmatics Interface 28), 275–295. Leiden: Brill.

Tao, Lian. 1998. *Shì-bù-shì* wenju shuo lue [On *shi-bu-shi* questions]. *Zhongguo Yuwen [Studies of the Chinese Language]* 2. 105–107.

Theiler, Nadine, Floris Roelofsen & Maria Aloni. 2019. Picky predicates: Why *believe* doesn't like interrogative complements, and other puzzles. *Natural Language Semantics* 27(2). 95–134. `https://doi.org/10.1007/s11050-019-09152-9`.

van Rooy, Robert & Marie Šafářová. 2003. On polar questions. In Robert B. Young & Yuping Zhou (eds.), *Proceedings of the 13th Semantics and Linguistic Theory Conference*, 292–309. Ithaca, NY: CLC. `https://doi.org/10.3765/salt.v13i0.2887`.

Velleman, Dan Bridges et al. 2012. It-clefts are IT (Inquiry Terminating) constructions. In Anca Chereches (ed.), *Proceedings of the 22nd Semantics and Linguistic Theory Conference*, 441–460. Ithaca, NY: CLC. `https://doi.org/10.3765/salt.v22i0.2640`.

Wang, Yaxuan. 2024. Derive the biased reading of *A-not-A* questions in Mandarin. In Patrick Farrell (ed.), *Proceedings of the Linguistic Society of America*, vol. 9, 5666. Washington, DC: Linguistic Society of America. `https://doi.org/10.3765/plsa.v9i1.5666`.

Ye, Shumian. 2021a. From maximality to bias: Biased A-not-A questions in Mandarin Chinese. In Joseph Rhyne et al. (eds.), *Proceedings of the 30th Semantics and Linguistic*

Theory Conference, 355–375. Ithaca, NY: CLC. https://doi.org/10.3765/salt.v30i0.4826.

Ye, Shumian. 2021b. *Xiandai hanyu pianxiang wenju de yiyi he jiegou [The meaning and structure of biased polar questions in Mandarin Chinese]*. Peking University dissertation.